# LECTURE SHEET

## EXERCISE-I

***(Types of reactions, chemical equilibria, law of mass action, equilibrium constant, characteristics of equilibrium constant, predicting the direction of the reaction, $K_p$, $K_c$ relation)***

### LEVEL-I (MAIN)

***Straight Objective Type Questions***

1. An example of an irreversible reaction
   1) $CH_3COOC_2H_5 + H_2O \rightarrow CH_3COOH + C_2H_5OH$
   2) $N_2 + O_2 \rightarrow 2NO$
   3) $NH_4HS \rightarrow NH_3 + H_2S$
   4) $BaCl_{2(aq)} + K_2SO_{4(aq)} \rightarrow BaSO_{4(s)} + 2KCl_{(aq)}$

2. $Fe^{+3}_{(aq)} + SCN^-_{(aq)} \rightleftharpoons [Fe(SCN)]^{+2}_{(aq)}$ is an example of
   1) Heterogeneous equilibrium
   2) Homogeneous equilibrium
   3) Reversible process that never attains equilibrium state
   4) Irreversible process that attains equilibrium state

3. When the rate of formation of reactants is equal to the rate of formation of products, this is known as,
   1) Chemical reaction
   2) Chemical equilibrium
   3) Chemical kinetics
   4) Chemical energetics

4. When $H_2$ and $I_2$ are mixed and equilibrium is attained, then
1) Amount of HI formed is equal to the amount of $H_2$ dissociated
2) HI dissociation stops
3) the reaction stops completely
4) Both forward and backward reactions proceed with same rate

5. At low temperature, Nitrogen dioxide, a reddish brown gas gets associated to form the colourless dinitrogen tetroxide as in the reaction $2NO_{2(g)} \rightleftharpoons N_2O_{4(g)}$. Then at equilibrium
1) There would be an increase in colour intensity
2) The mixture would become colourless
3) There would be a decrease in colour intensity
4) There would be no change in colour intensity

6. Assertion (A) : Introduction of catalyst does not affect position of equlibrium.
Reason (R) : For a reversible reaction, presence of a catalyst influences both forward & backward reaction reacts to same extent.
1) both A & R are true, R is the correct explanation of A
2) both A & R are true, R is not correct explanation of A
3) A is true, R is false
4) A is false, R is true

7. Law of mass action is applicable to
1) Homogeneous chemical equilibrium only
2) Heterogeneous chemical equilibrium only
3) Both homogeneous and Heterogeneous chemical equilibria
4) Physical equilibrium

8. Law of mass action is not applicable to $C_{(graphite)} \rightleftharpoons C_{(diamond)}$ because
1) It is a physical equilibrium
2) The process is not spontaneous
3) The process spontaneous
4) Both forms are crystalline

9. Active mass of 5.6 lit $N_2$ at STP
1) 22.4M
2) 0.25M
3) $\frac{1}{22.4}$M
4) 4M

10. What is the equilibrium expression for the reaction, $P_{4(s)} + 5O_{2(g)} \rightleftharpoons P_4O_{10(s)}$?
1) $K_c = \frac{[P_4O_{10}]}{[P_4][O_2]^5}$
2) $K_c = \frac{1}{[O_2]^5}$
3) $K_c = [O_2]^5$
4) $K_c = \frac{[P_4O_{10}]}{5[P_4][O_2]}$

11. As per law of mass action, for $NH_4HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$ ratio of rate constants of forward ($K_f$) & backward ($K_b$) reactions at equilibrium equals to
1) $[NH_4HS]$
2) $P_{NH_3} + P_{H_2S}$
3) $[H_2S] + [NH_3]$
4) $[NH_3][H_2S]$

12. Units of $K_c$ for $xA_{(g)} \rightleftharpoons yB_{(g)}$ is lit$^2$-mol$^{-2}$, then the values of x & y cannot be
1) 1, 2
2) 3, 2
3) 2, 3
4) All

13. The following are some statements about units of $K_c$ and $K_p$.
A) $K_p$ always has units.
B) $K_c$ has no units at all times.
C) If $\Delta n_g = 0$, then $K_p$ and $K_c$ have no units.
The correct set is
1) A and B
2) C only
3) C and A
4) A, B, C

14. $K_C$ for $N_2 + O_2 \rightleftharpoons 2NO$ is 'X', then for $NO \rightleftharpoons 1/2N_2 + 1/2\ O_2$, it is

1) $X^2$   2) $\sqrt{X}$   3) $\frac{1}{\sqrt{X}}$   4) $\frac{1}{X^2}$

15. For the hypothetical reactions, the equilibrium constant (K) values are given

$A \rightleftharpoons B;\ K_1 = 2.0$   $B \rightleftharpoons C;\ K_2 = 4.0$   $C \rightleftharpoons D;\ K_3 = 3.0$

The equilibrium constant for the reaction $A \rightleftharpoons D$ is

1) 48   2) 6   3) 24   4) 12

16. For which of the following reactions, $K_p\,(RT)^2 = K_c$

1) $PCl_{5(g)} \Leftrightarrow PCl_{3(g)} + Cl_{2(g)}$   2) $N_{2(g)} + 3H_{2(g)} \Leftrightarrow 2NH_{3(g)}$

3) $2SO_{2(g)} + O_{2(g)} \Leftrightarrow 2SO_{3(g)}$   4) $H_{2(g)} + I_{2(g)} \Leftrightarrow 2HI_{(g)}$

17. $K_p/K_c$ for $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (gaseous phase) at 400 K is

1) 400 R   2) $(400R)^2$   3) $(400R)^{-2}$   4) $(127)^{-2}$

18. The reaction $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$ is carried out in a 1 litre flask. If the same reaction is carried out in a 2 litre flask at the same temperature, the equilibrium constant will be

1) same   2) doubled   3) halved   4) decreased

19. A catalyst

1) Alters the equilibrium constant

2) Increases the equilibrium concentration of products

3) helps establishing the equilibrium quickly

4) Supplies energy to the reactants

20. If $N_2 + 3H_2 \rightleftharpoons 2NH_3$...(I) &

$N_2 + 3H_2 \xrightleftharpoons{Fe} 2NH_3$.... (II) are in equilibrium at same temperature, then

1) $K_c$ of I = $K_c$ of II   2) $K_c$ of I = $K_p$ of II   3) $K_c$ of I < $K_c$ of II   4) $K_p$ of II > $K_p$ of I

21. With increase in temperature, the value of equilibrium constant

1) Increases   2) Decreases

3) May increase or decrease   4) Remains constant

## LEVEL-II (ADVANCED)

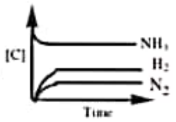
### *Straight Objective Type Questions*

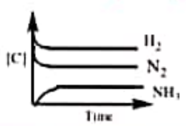
1. Attainment of "equilibrium state" with the help of "constancy in intensity of colour" is noticed in the case of _____ in a closed vessel

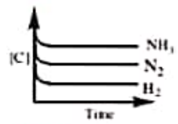
a) Decomposition of $CaCO_3$   b) Reaction between $N_2$ & $O_2$

c) Reaction between $H_2$ & $I_2$   d) Decomposition of $PCl_5$

2. Which of the following is correct for $N_2 + 3H_2 \rightleftharpoons 2NH_3$

a) 


b) 


c) 


d) All

3. Which one of the following has greater active mass

a) 200 gm of lime stone in 2 lit vessel
b) 90 gm of $CS_2$ liquid in 100 ml vessel
c) 56 gm of $N_2$ gas in 0.5 lit vessel
d) 1 mole of $O_2$ gas at STP

4. The equilibrium constants for the stepwise formation of MCl, $MCl_2$ and $MCl_3$ are a, b and c respectively. If the equilibrium contant of formation of $MCl_3$ is K, which of the following is correct?

a) $K = a + b + c$
b) $\frac{1}{K} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$
c) $\log K = \log a + \log b + \log c$
d) $K = \frac{1}{a} \times \frac{1}{b} \times \frac{1}{c}$

5. Sulphide ion in alkaline solution reacts with solid sulphur to form polysulphide ions having formulae $S_2^{2-}, S_3^{2-}, S_4^{2-}$ and so on. The equilibrium constant for the formation of $S_2^{2-}$ is 12($K_1$) and for the formation of $S_3^{2-}$ is 132 ($K_2$), both from S and $S^{2-}$. What is the equilibrium constant for the formation of $S_3^{2-}$ from $S_2^{2-}$ and S ?

a) 11
b) 12
c) 132
d) None of these

6. A vessel (A) contains 1 mole each of $N_2$ & $O_2$ and another vessel (B) contains 2 mole each of $N_2$ & $O_2$. Both vessels are heated to same temperature till equilibrium is established in both cases. Then, correct statement is

a) $K_c$ for $N_2 + O_2 \rightleftharpoons 2NO$ in A & B are in the ratio 1:2
b) $K_p$ for $N_2 + O_2 \rightleftharpoons 2NO$ in A & B are in the ratio 1:2
c) $K_c$ for $N_2 + O_2 \rightleftharpoons 2NO$ in A & B are equal
d) $K_p$ for $N_2 + O_2 \rightleftharpoons 2NO$ in A & B are in the ratio 2:1

7. For the equilibrium

$$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}; K_c = \frac{\alpha^2}{(1-\alpha)V}$$

temperature remaining constant.

a) $K_c$ will increase with increase in volume
b) $K_c$ will increase with decrease in volume
c) $K_c$ will not change with the change in volume
d) $K_c$ may increase or decrease with the change in volume depending upon its numerical value

8. The following are some statements about equilibrium constant

A) The value of K is affected by temperature
B) The equilibrium constant gives idea about the extent of completion of reaction
C) The equilibrium constant is affected by volume and pressure

The correct combination is

a) A and B
b) B and C
c) C and A
d) all

9. 1) $N_2 + 2O_2 \Leftrightarrow 2NO_2 ; K_c = 2 \times 10^{-31}$
2) $2NO \Leftrightarrow 2N_2 + O_2 ; K_c = 2.2 \times 10^{-33}$
3) $2N_2O_5 \Leftrightarrow 2N_2 + 5O_2 ; K_c = 3.8 \times 10^{-32}$
4) $2N_2 + O_2 \Leftrightarrow 2N_2O ; K_c = 4 \times 10^{-32}$

From the above data, the most stable oxide is

a) $NO_2$
b) NO
c) $N_2O_5$
d) $N_2O$

10. Consider the reaction $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$ for which $K_c = 278\ M^{-1}$. 0.001 mole of each of the reagents $SO_{2(g)}$, $O_{2(g)}$ and $SO_{3(g)}$ are mixed in a 1.0$l$ flask. Determine the reaction quotient of the system and the spontaneous direction of the system :

a) $Q_c = 1000$; the equilibrium shifts to the right

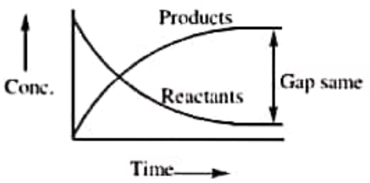
b) $Q_c = 1000$; the equilibrium shifts to the left

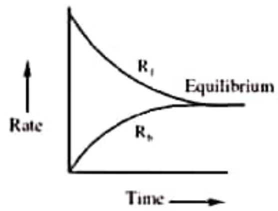
c) $Q_c = 0.001$; the equilibrium shifts to the left

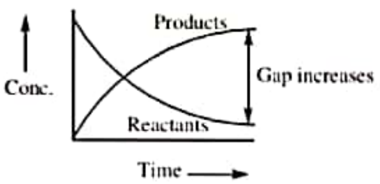
d) $Q_c = 0.001$; the equilibrium shifts to the right

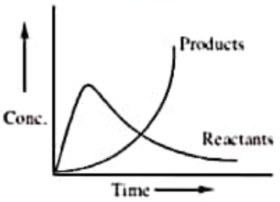
## More than One correct answer Type Questions

11. For the reaction $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$. Which of the graph is/are correct?

a) 


b) 


c) 


d) 


12. In which of the following equilibrium, the value of $K_p$ is less than $K_C$?

a) $N_2O_4 \rightleftharpoons 2NO_2$ b) $N_2 + O_2 \rightleftharpoons 2NO$ c) $N_2 + 3H_2 \rightleftharpoons 2NH_3$ d) $2SO_2 + O_2 \rightleftharpoons 2SO_3$

13. The equilibrium constant of a chemical reaction

a) is constant at a given temperature

b) has only numerical value and carries no units

c) its value always depends upon the units in which the concentrations of species involved in chemical reaction are measured

d) changes if the stoichiometric coefficients of all the species involved in the chemical equation are changed

14. Select the correct statements.

a) The value of $Q_c$ for a given reaction is constant

b) A change in the physical state of gaseous product shows a change in the value of equilibrium constant of that reaction

c) The value of ratio of $\frac{Q_c}{K_c}$ can be used to predict the direction in which a system will proceed spontaneously towards equilibrium

d) The rate constants for forward and backward reaction for a reversible reaction always increases with temperature but their ratio $\left(\text{i.e } \frac{k_f}{k_b}\right)$ may increase or decrease with temperature.

## Linked Comprehension Type Questions

***Passage :***

Mass action ratio or reaction quotient Q for a reaction can be calculated using the law of mass action $A(g)+B(g)\rightleftharpoons C(g)+D(g)$ $Q=\frac{[C][D]}{[A][B]}$. The value of Q decides whether the reaction is at equilibrium or not. At equilibrium, Q = K. For non equilibrium process, $Q \neq K$. When Q > K, reaction will favour backward direction and when Q < K, it will favour forward direction.

15. The reaction quotient Q for : $N_{2(g)}+3H_{2(g)}\rightleftharpoons 2NH_{3(g)}$ is given by $Q=\frac{[NH_3]^2}{[N_2][H_2]^3}$. The reaction will proceed in backward direction, when :

a) $Q=K_c$ b) $Q<K_c$ c) $Q>K_c$ d) $Q=0$

16. For the reaction: $2A+B\rightleftharpoons 3C$ at 298 K, $K_c=49$

A 3L vessel contains 2,1 and 3 moles of A, B and C respectively.

The reaction at the same temperature:

a) must proceed in forward direction
b) must proceed in backward direction
c) must be in equilibrium
d) cannot be predicted

17. In a reaction mixture containing $H_2$, $N_2$ and $NH_3$ at partial pressure of 2 atm, 1 atm and 3 atm respectively, the value of $K_p$ at 725 K is $4.28\times10^{-5}atm^{-2}$. In which direction will the reaction $N_{2(g)}+3H_{2(g)}\rightleftharpoons 2NH_{3(g)}$ proceed?

a) Forward
b) Backward
c) No net reaction
d) Direction cannot be predicted

## Matrix Matching Type Questions

18.

| Column-I (Reaction) | Column-II $\left(\frac{K_p}{K_c}\right)$ |
|---|---|
| A) $A_{2(g)}+3B_{2(g)}\rightleftharpoons 2AB_{3(g)}$ | P) $(RT)^{-2}$ |
| B) $A_{2(g)}+B_{2(g)}\rightleftharpoons 2AB_{(g)}$ | Q) $(RT)^0$ |
| C) $A_{(s)}+\frac{3}{2}B_{2(g)}\rightleftharpoons AB_{3(g)}$ | R) $(RT)^{1/2}$ |
| D) $AB_{2(g)}\rightleftharpoons AB_{(g)}+\frac{1}{2}B_{2(g)}$ | S) $(RT)^{-1/2}$ |

19.

| Column-I | Column-II |
|---|---|
| A) Q = K | P) Reaction is nearer to completion |
| B) Q < K | Q) Reaction is not at equilibrium |
| C) Q > K | R) Reaction is faster in forward direction |
| D) K >>> 1 | S) Reaction at equilibrium |

## EXERCISE-II

***($K_p$, $K_c$ numerical calculations, Relation between degree of dissociation ($\alpha$) and vapour density)***

### LEVEL-I (MAIN)

***Straight Objective Type Questions***

1. Starting from 'a' moles of $H_2$ and 'b' moles of $I_2$ an equilibrium $H_2 + I_2 \Leftrightarrow 2HI$ is established with 2x moles of HI. The equilibrium constant $K_C$ is

   1) $\frac{4x^2}{ab}$  2) $\frac{4x^2}{(a-x)(b-x)}$  3) $\frac{2x}{(a-x)(b-x)}$  4) $\frac{4x^2}{(a-2x)(b-2x)}$

2. An equilibrium mixture for the reaction, $2H_2S_{(g)} \rightleftharpoons 2H_{2(g)} + S_{2(g)}$ has 1 mole of $H_2S$, 0.2 mole of $H_2$ and 0.8 mole of $S_2$ in 2 L flask. The value of $K_c$ in mol $L^{-1}$ is

   1) 0.004  2) 0.08  3) 0.016  4) 0.16

3. The equilibrium constant for the reaction $H_{2(g)} + I_{2(g)} \Leftrightarrow 2HI_{(g)}$ is 2 at a certain temperature. The equilibrium concentrations of $H_2$ and HI are 2 mole.lit$^{-1}$. What is the equilibrium concentration (in mole.lit$^{-1}$) of $I_2$?

   1) 16  2) 4  3) 8  4) 1

4. 1.2 moles of $SO_3$ are allowed to dissociate in a 2 litre vessel the reaction is $2SO_{3(g)} \Leftrightarrow 2SO_{2(g)} + O_{2(g)}$ and the concentration of oxygen at equilibrium is 0.1 mole per litre. The total number of moles at equilibrium will be

   1) 2  2) 1.4  3) 0.8  4) 1.6

5. A mixture of 2 moles of $N_2$ and 8 moles of $H_2$ are heated in a 2 lit vessel, till equilibrium is established. At equilibrium, 0.4 moles of $N_2$ was present. The equilibrium concentration of $H_2$ will be

   1) 2 mole/lit  2) 4 mole/lit  3) 1.6 mole/lit  4) 1 mole/lit

6. 1.0 mole of ethyl alcohol and 1.0 mole of acetic acid are mixed. At equilibrium, 0.666 mole of ester is formed. The value of equilibrium constant is

   1) 1/4  2) 1/2  3) 4  4) 3

7. The reaction $N_2 + 3H_2 \Leftrightarrow 2NH_3$ takes place at 450$^0$C. 1 mole of $N_2$ and 2 mole of $H_2$ are mixed in a 1 litre vessel and 1mole of $NH_3$ is formed at equilibrium. Then $K_c$ for the above reaction is

   1) 4  2) 8  3) 16  4) 32

8. 1 mole $CaCO_{3(s)}$ is heated in 11.2 lit vessel so that equilibrium is established at 819 K. If $K_p$ for $CaCO_3 \rightleftharpoons CaO + CO_2$ at this temperature is 2 atm, equilibrium concentration of $CO_2$ (in mol-lit$^{-1}$)

   1) 1/3  2) 1/11.2  3) 1/33.6  4) 1/22.4

9. At a certain temperature the equilibrium constant $K_c$ is 0.25 for the reaction $A_{2(g)} + B_{2(g)} \rightleftharpoons C_{2(g)} + D_{2(g)}$. If we take 1 mole of each of the four gases in a 10 litre container, what would be equilibrium concentration of $A_{2(g)}$?

   1) 0.331 M  2) 0.033 M  3) 0.133M  4) 1.33 M

10. The compounds A and B are mixed in equimolar proportion to from the products, $A + B \rightleftharpoons C + D$. At equilibrium, one third of A and B are consumed. The equilibrium constant for the reaction is

1) 0.5 2) 4.0 3) 2.5 4) 0.25

11. For which of the following reactions, the degree of dissociation ( $\alpha$ ) and equilibrium constant ($K_p$) are related as $K_p = \dfrac{4\alpha^2 p}{(1-\alpha^2)}$

1) $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$
2) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$
3) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$
4) $PCl_{3(g)} + Cl_{2(g)} \rightleftharpoons PCl_{5(g)}$

12. 20 gm of $CaCO_3$ is allowed to dissociate in a 5.6 litres container at $819^0$ C. If 50% of $CaCO_3$ is dissociated at equilibrium the '$K_p$' value is

1) 5 atm 2) 1.6 atm 3) 4.8 atm 4) 10 atm

13. $N_2O_4$ at an initial pressure of 2 atm and 300K dissociates to an extent of 20% at the same temperature by the time equilibrium is established. $K_p$ for the reaction $2NO_2 \rightleftharpoons N_2O_4$ is

1) 0.4 2) 0.8 3) 2.5 4) 1.6

14. For the reaction $N_2O_4 \rightleftharpoons 2NO_{2(g)}$, the degree of dissociation at equilibrium is 0.2 at 1 atm. Then $K_p$ will be

1) 1/2 2) 1/4 3) 1/6 4) 1/8

## LEVEL-II (ADVANCED)

### *Straight Objective Type Questions*

1. For the gaseous phase reaction $2A + B \rightleftharpoons 2C + D$, initially there are 2 mole each of A & B. If 0.4 mol of D is present at equilibrium at a given T & P, in-correct relationship is

a) $P_A < P_B$ & $P_D < P_C$
b) $P_A = P_C$ & $P_B = P_D$
c) $P_C = 2P_D$ & $P_A = 3P_B/4$
d) $P_A > P_D$ & $P_B > P_C$

2. A vessel at 1000 K contains $CO_2$ with a pressure of 0.5 atm. Some of $CO_2$ is converted into CO on addtion of graphite. The value of 'K' at equilibrium when total pressure is 0.8 atm will be

a) 0.18 atm
b) 1.8 atm
c) 2 atm
d) 1 atm

3. Initially 0.8 mole of $PCl_5$ and 0.2 mole of $PCl_3$ are mixed in one litre vessel. At equilibrium 0.4 mole of $PCl_3$ is present. The value of $K_C$ for the reaction $PCl_{5(g)} \Leftrightarrow PCl_{3(g)} + Cl_{2(g)}$ is

a) $0.13 molL^{-1}$
b) $0.05 molL^{-1}$
c) $0.065 molL^{-1}$
d) $0.1 molL^{-1}$

4. A vessel contains 1 mole $PCl_5$ (g) at 4 atm and 0.5 mole $PCl_3$ is formed at equilibrium. Now, equilibrium pressure of mixture is (assume ideal behavior)

a) 16 atm b) 6 atm c) 2 atm d) 4.5 atm

5. A mixture of Nitrogen and Hydrogen (1:3 mole ratio) is at an initial pressure of 200atm. If 20% of the mixture reacts by the time equilibrium is reached, the equilibrium pressure of the mixture is

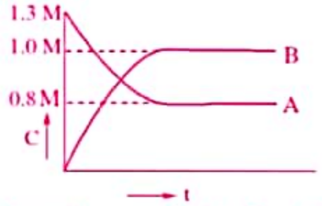
a) Data insufficient b) 180 atm c) 170 atm d) 160 atm

6. Vapour density of $N_2O_4$ at 60°C is found to be 30.6. The degree of dissociation of $N_2O_4$ is :

a) 10% b) 20% c) 40% d) 50%

7. In a 500 ml flask, the degree of dissociation of $PCl_5$ at equilibrium is 40% and the initial amount is 5 moles. The value of equilibrium constant in mole lit$^{-1}$ for the decomposition of $PCl_5$ is

a) 1.33 b) 2.66 c) 5.32 d) 4.66

8. 


As per the graph what is $K_c$ of $A \rightleftharpoons nB$

a) 1.25 M b) 1.48 M c) 2.16 M d) 0.8 M

9. The equilibrium constant $K_p$ (in atm) for the reaction is 9 at 7 atm and 300 K. $A_{2(g)} \rightleftharpoons B_{2(g)} + C_{2(g)}$. calcuate the average molar mass (in gm/mol) of an equilibrium mixture.

Given : Molar mass of $A_2$, $B_2$ and $C_2$ are 70, 49 and 21 gm/mol respectively.

a) 50 b) 45 c) 40 d) 37.5

## More than One correct answer Type Questions

10. In an equilibrium $A + B \rightleftharpoons C + D$ ; A and B are mixed in a vessel at temperature T. The initial concentration of A was twice the initial concentration of B. After the equilibrium has reached, the concentration of C was thrice the equilibrium concentration of B, then

a) $K_c = 1.8$ b) $[A]_{eqm} = [D]_{eqm} \times 1.66$

c) $K_c = 2.8$ d) $[A]_{eqm} = [C]_{eqm} \times 2.5$

11. $AB_{2(g)}$ dissociates as $AB_{2(g)} \rightleftharpoons AB_{(g)} + B_{(g)}$. The initial pressure of $AB_2$ is 600 mm Hg and equilibrium pressure of the mixture of gases is 800 mm Hg. Then

a) $\alpha = 33.3\%$

b) $K_p$=100mm

c) equilibrium pressure of $AB_2$ is 400mm

d) equilibrium pressure of B is 200mm

12. One mole of $N_2O_{4(g)}$ at 300 K is kept in a closed container under one atmosphere. It is heated to 600 K to decompose to $NO_{2(g)}$. If the resultant pressure at equilibrium is 2.4 atm, then which is correct?

a) % dissociation = 20%

b) pressure of $NO_2$ at equilibrium = 0.8 atm

c) pressure of $N_2O_4$ at equilibrium = 1.2 atm

d) $K_p = 0.4$ atm

13. $2O_3 \rightleftharpoons 3O_2$. Ozone dissociates until the molecular weight of the equilibrium mixture becomes 40. Initial pressure of $O_3$ is 600 mm Hg. Then

a) $K_p = 360$ mm
b) degree of dissociation = 0.4
c) $P_{eq} = 720$ mm
d) $K_p = P_{O_3} = P_{O_2}$

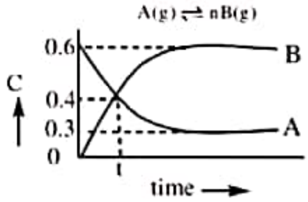
14. For the reaction $AB_{2(g)} \rightleftharpoons AB_{(g)} + B_{(g)}$ if $\alpha$ is negligiable w.r.t 1 then degree of dissociation ($\alpha$) of $AB_2$ is proportional to

a) $\frac{1}{p}$
b) $\frac{1}{v}$
c) $\frac{1}{\sqrt{p}}$
d) $\sqrt{v}$

## Linked Comprehension Type Questions

***Passage :***

The progress of the reaction $A \rightleftharpoons nB$ with time t is shown in Fig.
From this information answer the following questions




15. The value of n is

a) 1
b) 2
c) 3
d) 4

16. The equilibrium constant $K_c$ is

a) 1.2
b) 1.4
c) 1.6
d) 1.8

17. At time 't'

a) $r_f > r_b$
b) $r_f < r_b$
c) $r_f = r_b$
d) $k_f = k_b$

## Integer Type Questions

18. 4.5 moles each of hydrogen and iodine are heated in a sealed 10 L vessel. At equilibrium 3 moles of hydrogen iodide was found. Find the equilibrium constant for $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$.

19. For the equilibrium of the reaction $NH_4Cl_{(s)} \rightleftharpoons NH_{3(g)} + HCl_{(g)}, K_p = 81 atm^2$. Total pressure at equilibrium will be x times the pressure of $NH_3$. The value of x will be ____

20. For the reaction in equilibrium : $2NOBr_{(g)} \rightleftharpoons 2NO_{(g)} + Br_{2(g)}$ if $P_{Br_2}$ is $\frac{P}{9}$ at equilibrium and P is total pressure. If $\frac{P}{Kp} = x^2$ then what is x value.

21. $N_2O_4 \rightleftharpoons 2NO_2$. At 300K $N_2O_4$ at 6 atmosphere is heated to 600K where equilibrium is established with a total pressure of 16 atmospheres. What is $K_p$ at 600K?

22. In a reaction at equilibrium, 'X' mole of the reactant A decompose to give 1 mole each of C and D. It has been found that the fraction of A decomposed at equilibrium is independent of initial concentration of A. Calculate X.

## EXERCISE-III

*(Factors affecting equilibria, Le-chatelier's principle, Thermodynamic relationship)*

### LEVEL-I (MAIN)

*Straight Objective Type Questions*

1. Le chatelier's principle is applicable to
   1) Chemical equilibria only 2) Physical equilibria only
   3) Both physical and chemical equilibria 4) Gaseous systems only

2. For the Chemical reaction $A_{2(g)} + B_{2(g)} \rightleftharpoons 2AB_{(g)}$ the amount of AB at equilibrium is affected by
   1) Temperature and pressure 2) Temperature only
   3) Pressure only 4) Temperature, pressure and Catalyst

3. Increase of pressure favours the forward reaction in the following equilibrium
   1) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$ 2) $2NO_{2(g)} \rightleftharpoons N_2O_{4(g)}$
   3) $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ 4) $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$

4. Exothermic formation represented by equation $Cl_{2(g)} + 3F_{2(g)} \rightleftharpoons 2ClF_{3(g)}$, $\Delta H = -339$ KJ. Which of the following will increase the quantity of $ClF_3$ in equilibrium mixture ?
   1) Increase in temperature 2) Removing $Cl_2$
   3) Increasing volume of vessel 4) Adding $F_2$

5. The yield of product in the reaction $2A_{(g)} + B_{(g)} \rightleftharpoons 2C_{(g)} + Q$ KJ would be lower at :
   1) low temperature and low pressure 2) high temperature and high pressure
   3) low temperature and too high pressure 4) high temperature and low pressure

6. Under what conditions of temperature and pressure the formation of atomic hydrogen from molecular hydrogen will be favoured.
   1) High temperature and high pressure 2) Low temperature and low pressure
   3) High temperature and low pressure 4) Low temperature and high pressure

7. Consider the reaction equilibrium, $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}; \Delta H^o = -198$ kJ . On the basis of Le Chatelier's principle, the condition favourable for the forward reaction is
   1) Lowering of temperature as well as pressure 2) Increase of temperature as well as pressure
   3) Lowering of temperature and increase of pressure 4) Any value of temperature and pressure

8. The following are some statements regarding dissociation of lime stone according to the equation $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$; $\Delta H = 110$ k.J. The reaction is carried in a closed vessel.
   A) The pressure of $CO_2$ increases when temperature is increased.
   B) The pressure of $CO_2$ increases when temperature is decreased
   C) The pressure of $CO_2$ increases when amount of $CaCO_3$ is decreased.
   The incorrect statements are.
   1) A and B 2) B and C 3) A and C 4) All A, B, C

9. For the physical equilibrium ice $\Leftrightarrow$ water, the forward reaction is not favoured by
   1) Increasing pressure 2) Increasing temperature
   3) Keeping in contact with hot water 4) Taking more ice

10. Assertion (A) : The degree of decomposition of $PCl_5$ is more at low pressures.
Reason (R) : In a reversible reaction, on increasing the pressure, the equilibrium shifts in the direction in which decrease in volume takes place.
1) both A & R are true, R is the correct explanation of A
2) both A & R are true, R is not correct explanation of A
3) A is true, R is false
4) A is false, R is true

11. At equilibrium state
1) $\Delta G^0 = 0$
2) $\Delta G$=negative
3) $\Delta G$=zero
4) $\Delta G$=positive

12. A plot of Gibbs energy of a reaction mixture against the extent of the reaction is :
1) minimum at equilibrium
2) zero at equilibrium
3) maximum at equilibrium
4) None of these

13. If $\Delta G^0$ for the reaction given below is 1.7 kJ; the equilibrium constant for a reaction.
$2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)}$ at $25^0C$ is :
1) 24.0
2) 3.9
3) 2.0
4) 0.5

14. The intercepts and slope of graph of log $K_p^o$ Vs $\frac{1}{T}$ are :
1) $\Delta G^o, \Delta H^o$
2) $\frac{\Delta G^o}{2.303\,R}, \frac{\Delta H^o}{2.303\,R}$
3) $\frac{\Delta S^o}{2.303R}, \frac{-\Delta H^o}{2.303R}$
4) $\frac{\Delta H^o}{2.3003\,R}, \Delta S^o$

## LEVEL-II (ADVANCED)

### Straight Objective Type Questions

1. For a given reaction $K_p < K_c$. Increase of pressure favours
a) the backward reaction
b) no reaction
c) the forward reaction
d) both forward and backward reactions equally

2. $K_c$ value of a gaseous reaction is 5mole / lit. If pressure is increased
a) Forward reaction is favoured
b) Backward reaction is favoured
c) Reaction is uneffected
d) $K_c$ value increases

3. Consider the following equilibrium
$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ in a closed container. At a fixed temperature, the volume of the reaction container is halved. For this change, which of the following statements holds true regarding the equilibrium constant ($K_p$) and degree of dissociation ($\alpha$) ?
a) Neither $K_p$ nor $\alpha$ changes
b) Both $K_p$ and $\alpha$ change
c) $K_p$ changes, but $\alpha$ does not change
d) $K_p$ does not change, but $\alpha$ changes

4. Densities of diamond and graphite are 3.5 and 2.5 gm/ml. $C_{(diamond)} \rightleftharpoons C_{(graphite)}$ : $\Delta_r H = -1.9$ KJ / mole favourable conditions for formation of diamond are
a) high pressure and low temperature
b) low pressure and high temperature
c) high pressure and high temperature
d) low pressure and low temperature

5. Following two eqilibrium are simultaneously establised in a container $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$, $CO_{(g)} + Cl_{2(g)} \rightleftharpoons COCl_{2(g)}$. If some $Ni_{(s)}$ is introduced in the container forming $Ni(CO)_4$ then at new equilibrium

a) $PCl_3$ concentration will increase
b) $PCl_3$ concentration will decrease
c) $Cl_2$ concentration will remain same
d) CO concentration will remain same

6. Addition of 1 mole of $N_2$ is made to an equilibrium mixture of $PCl_5$ in a piston fitted cylinder. Which of the following is correct.

a) No effect on equilibrium constant $K_c$ or $K_p$
b) The degree of dissociation of $PCl_5$ decreases
c) The volume of container does not change
d) No effect on equilibrium concentrations

7. $A_{(s)} + B_{(g)} + \text{heat} \rightleftharpoons 2C_{(s)} + 2D_{(g)}$. At equilibrium the pressure of 'B' is doubled. By what factor the concentration of 'D' should change, to reattain the equilibrium

a) $\sqrt{2}$ b) 2 c) 3 d) $\sqrt{3}$

8. $PCl_5$ dissociates to 25% in a three litre flask. To what volume the container must be changed such that $PCl_5$ dissociates to 50% extent

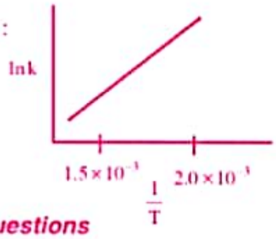
a) 18 b) 12 c) 6 d) 24

9. Following reaction in equilibrium at $25^0C$:

$2NO(g, 1\times10^{-5}\text{ atm}) + Cl_2(g, 1\times10^{-2}\text{ atm}) \rightleftharpoons 2NOCl(g, 1\times10^{-2}\text{ atm})$ $\Delta G^0$ is

a) –45.65 kJ b) –28.53 kJ c) –22.82 KJ d) –57.06 KJ

10. The graph relates in $K_{eq}$ Vs $\frac{1}{T}$ for a reaction. The reaction must be :




a) exothermic
b) endothermic
c) $\Delta H$ is negligible
d) spontaneous

## More than One correct answer Type Questions

11. Consider the following reaction, $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ the forward reaction at constant temperature is favoured by

a) introducing an inert gas at constant volume
b) introducing $Cl_2$ gas at constant volume
c) introducing an inert gas at constant pressure
d) increasing the volume of the container

12. When $NaNO_3$ is heated in a closed vessel, $O_2$ is liberated and $NaNO_2$ is left behind.

At equilibrium: $NaNO_{3(s)} \rightleftharpoons NaNO_{2(s)} + \frac{1}{2}O_{2(g)}$

a) addition of $NaNO_2$ favours reverse reaction
b) addition of $NaNO_3$ favours forward reaction
c) increasing temperature favours forward reaction
d) increasing pressure reduces the rate of forward reaction

13. In what manner the increase of pressure will affect the equation. $C_{(s)} + H_2O_{(g)} \rightleftharpoons CO_{(g)} + H_{2(g)}$

a) Shift in the forward direction
b) Shift in the reverse reaction
c) Increase in the yield of $H_2$
d) from high moles to low gaseous moles

14. The pressure of an equilibrium mixture of the three gases NO, $Cl_2$ and NOCl $2NO_{(g)} + Cl_{2(g)} \rightleftharpoons 2NOCl_{(g)}$ is suddenly decreased by doubling the volume of the container at constant temperature. When the system returns to equilibrium :
a) the concentration of NOCl will be increased
b) the value of the equilibrium constant $K_c$ will be increased
c) the number of moles of $Cl_2$ will be increased
d) the number of moles of NOCl will be decreased

15. Pure ammonia is placed in a vessel at a temperature where its degree of dissociation ($\alpha$) is appreciable. At equilibrium $2NH_3 \rightleftharpoons N_2 + 3H_2$.
a) $K_p$ does not change with pressure
b) $\alpha$ changes with pressure
c) concentration of $NH_3$ does not change with pressure
d) concentration of $H_2$ is more than that of $N_2$

16. For a physical equilibrium, $H_2O$ (Ice) $\rightleftharpoons$ $H_2O$(Water) which of the following is the true statement :
a) The pressure change do not affect the equilibrium
b) More of ice melts, if pressure on the system is increased
c) More of liquid freezes, if pressure on the system is increased
d) More of ice melts, if temperature of the system is increased

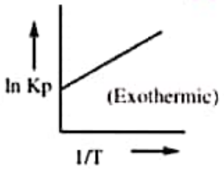
17. Two systems, $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ and $COCl_{2(g)} \rightleftharpoons CO_{(g)} + Cl_{2(g)}$ are simultaneously in equilibrium in a vessel at constant volume. If some CO is introduced into the vessel, then at the new equilibrium, the concentration of :
a) $PCl_5$ is greater than that before
b) $PCl_3$ is greater than that before
c) $PCl_5$ is less than that before
d) $COCl_2$ is more than that before

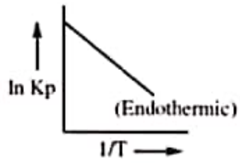
18. When a sample of $NO_2$ is placed in a container, the equilibrium is rapidly established. If the equilibrium mixture is of a darker colour at high temperatures and at low pressures, which of the following statements about the reaction is true ? $2NO_{2(g)} \rightleftharpoons N_2O_{4(g)}$
a) Formation of $N_2O_4$ from $NO_2$ is exothermic and $N_2O_4$ is darker in colour than $NO_2$
b) Formation of $N_2O_4$ from $NO_2$ is exothermic and $NO_2$ is darker in colour than $N_2O_4$
c) Rate of transformation of $NO_2$ to $N_2O_4$ is equal to the rate of transformation of $N_2O_4$ to $NO_2$, at equilibrium
d) There is no change in the concentrations, colour, vapour density of reaction mixture, at equilibrium

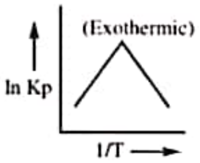
19. Under what conditions will $CuSO_4.5H_2O$ be efflorescent at $25^0C$? For the reaction
$CuSO_4.5H_2O_{(s)} \rightleftharpoons CuSO_4.3H_2O_{(s)} + 2H_2O_{(g)}$
$K_p$ at 298K is $1.086\times10^{-4}$ $atm^2$ and vapour pressure of water is 23.8 Torr.
a) If relative humidity < 33.3%, efflorescence takes place
b) If relative humidity > 33.3%, efflorescence takes place
c) The salt does not act as efflorescent at $25^0C$
d) The salt effloreses if the vapour pressure of water vapour in atmosphere < 7.92 mm.

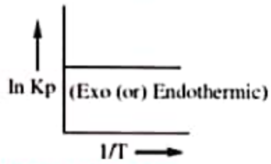
20. The equilibrium pressure of $NH_4CN_{(s)} \rightleftharpoons NH_{3(g)} + HCN_{(g)}$ is 4 atm. In an experiment, if $NH_4CN_{(s)}$ is allowed to decompose in the presence of $NH_3$ at 3 atm, then
a) pressure of HCN = 2atm (at old equilibrium state)
b) pressure of HCN = 1atm (at new equilibrium state)
c) $K_p = 4$
d) pressure of $NH_3$ = 4atm (at new equilibrium state)

21. Which of the following graphs is correct for a reaction?

a) 


b) 


c) 


d) 


## Linked Comprehension Type Questions

***Passage-I :***

As per Lechatlier's Principle any stress applied on the equilibrium state is minimised by shifting of equilibrium.

22. $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ ; $SO_2Cl_2 \rightleftharpoons SO_2 + Cl_2$

Both equilibria exist together in a flask. If some $SO_2$ is introduced into the flask ____

a) $SO_2Cl_2$ concentration decreases
b) degree of dissociation of $PCl_5$ decreases
c) degree of dissociation of $PCl_5$ increases
d) $PCl_3$ concentration decreases

23. In which cases introduction of inert gas shifts the equilibrium in forward direction at constant pressure ?

a) $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$
b) $N_2 + 3H_2 \rightleftharpoons 2NH_3$
c) $N_2O_4 \rightleftharpoons 2NO_2$
d) $S_{8(r)} \rightleftharpoons S_{8(m)}$

24. Increase in volume of container shifts the equilibrium in forward direction in the case of ____

a) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$
b) $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$
c) $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$
d) $C_2H_5OH_{(l)} + CH_3COOH_{(l)} \rightleftharpoons CH_3COOC_2H_{5(l)} + H_2O_{(l)}$

***Passage-II :***

AT 673 K, in the formation of $NH_3$ from $N_2$ and $H_2$, the partial pressures of $N_2$, $H_2$ and $NH_3$ at equilibrium are 0.5, 1 and $9 \times 10^{-3}$ atm respectively. ( $N_2 + 3H_2 \rightleftharpoons 2NH_3 + Q$ )

25. Calculate $K_c$ for the reaction.

a) 0.1 b) 0.2 c) 0.4 d) 0.5

26. Report $\Delta G^o$ using $K_P$ value

a) 22.66 kcal b) 11.33 kcal c) 61.33 kcal d) 30.2 kcal

27. Report $\Delta G^o$ using $K_C$ Value

a) 9.3 kcal b) 430 kcal c) 220 kcal d) 0.930 kcal

## Matrix Matching Type Questions

28. **Column-I (F.R. favourable)** — **Column-II**

| Column-I (F.R. favourable) | Column-II |
|---|---|
| A) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)} + Q$ | P) High temperature |
| B) $2SO_2 + O_{2(g)} \rightleftharpoons 2SO_{3(g)} + Q$ | Q) Low temperature |
| C) $2NH_{3(g)} \rightleftharpoons N_{2(g)} + 3H_{2(g)} - Q$ | R) High pressure |
| D) $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)} - Q$ | S) Low pressure |

29. **Column-I** — **Column-II**

| Column-I | Column-II |
|---|---|
| A) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$ | P) Unaffected by inert gas addition |
| B) $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$ | Q) Forward shift by rise in pressure and backward shift by inert gas addition at constant pressure |
| C) $N_{2(g)} + 3H_2 + (g) \rightleftharpoons 2NH_{3(g)}$ | R) Unaffected by increase in pressure |
| D) $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ | S) Backward shift by rise in pressure and forward shift by inert gas addition at constant pressure |

30. **Column-I** — **Column-II**

| Column-I | Column-II |
|---|---|
| A) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}; \Delta H = -ve$ | P) K increases with increase in temperature |
| B) $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}; \Delta H = +ve$ | Q) K decreases with increase in temperature |
| C) $A_{(g)} + B_{(g)} \rightleftharpoons 2C_{(g)} + D_{(g)}; \Delta H = +ve$ | R) Pressure has no effect |
| D) $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}; \Delta H = +ve$ | S) Product moles increase due to addition of inert gas at constant pressure |

## Integer Type Questions

31. For $A_{(g)} \rightleftharpoons B_{(g)} + C_{(g)}$, $K_p = 0.118$ atm at 1338K and $\Delta H = 42.4$KCal. Then equilibrium constant at 1405K is $x^2 \times 10^{-2}$. What is x?

32. At equilibrium, $K_p = 1$ then the value of $\Delta G^0$ will be equal to ____

## KEY SHEET (LECTURE SHEET)

### EXERCISE-I

**LEVEL-I**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1) 4 | 2) 2 | 3) 2 | 4) 4 | 5) 4 | 6) 1 | 7) 3 | 8) 1 |
| 9) 3 | 10) 2 | 11) 4 | 12) 4 | 13) 2 | 14) 3 | 15) 3 | 16) 2 |
| 17) 3 | 18) 1 | 19) 3 | 20) 1 | 21) 3 | | | |

**LEVEL-II**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1) c | 2) b | 3) c | 4) c | 5) a | 6) c | 7) c | 8) a |
| 9) b | 10) b | 11) ab | 12) cd | 13) acd | 14) bcd | 15) c | 16) a |

17) b 18) A-P; B-Q; C-S; D-R 19) A-S; B-QR; C-Q; D-P

## EXERCISE-II

**LEVEL-I**

1) 2 2) 3 3) 4 4) 2 5) 3 6) 3 7) 3 8) 3
9) 3 10) 4 11) 1 12) 2 13) 3 14) 3

**LEVEL-II**

1) b 2) b 3) a 4) b 5) b 6) d 7) b 8) a
9) c 10) ab 11) abcd 12) abd 13) abcd 14) cd 15) b 16) a
17) a 18) 1 19) 2 20) 9 21) 8 22) 2

## EXERCISE-III

**LEVEL-I**

1) 3 2) 2 3) 2 4) 4 5) 4 6) 3 7) 3 8) 2
9) 4 10) 1 11) 3 12) 1 13) 4 14) 3

**LEVEL-II**

1) c 2) b 3) d 4) c 5) b 6) a 7) a 8) a
9) a 10) a 11) cd 12) cd 13) bd 14) cd 15) abd 16) bd
17) bcd 18) bcd 19) ad 20) abcd 21) ab 22) c 23) c 24) b
25) d 26) b 27) d 28) A-Q; B-QR; C-PS, D-PS
29) A-PR; B-S; C-Q; D-S 30) A-Q; B-PR; C-PS; D-PS 31) 5 32) 0